

How do we get more food to feed the world? Historically, agriculture has alternated between two strategies: coax greater yield out of land already devoted to farming or find extra space to grow more. But how do we face the pending food crisis? I believe GM food is a weapon in the arsenal that we need. They have been grown for two decades on billions of hectares of land. And no one has ever been shown to have been affected negatively by eating GM food.

These are fears of politics, of corporatisation and globalisation, not fears about science. Science isn't a company, a country, or a policy. It is a process. We shouldn't get confused by the way a company uses science and the way science can do good. And anyone who plunges headlong into environmental and social changes without judging the risks and benefits is doing a disservice to humanity.

## "Giving too much credence to technology can be misguided"

**SUMAN SAHAI** CONVENER, GENE CAMPAIGN

The oft-repeated phrase that genetic engineering is very precise is not true. Unfortunately, the mythology of genetic engineering is replete with examples that are not substantiated by facts. Hunger happens when a person does not have access to productive assets, like land or water, to grow food, or does not have a job and enough money to buy food. We are not realising the genetic potential of our crops because farmers can't afford enough fertilisers and soil health cannot be maintained. Certainly, technology can play a role, but to give credence to it beyond what it has so far shown is perhaps misguided.

Bt brinjal has to be regulated cautiously because scientists have acknowledged that there are safety concerns. There were grave and outstanding questions about the way it had been regulated. India has a policy on mandatory labelling of GM foods. We gave permission for Bt brinjal's

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SAHAI, SESSION CHAIRPERSON  
RAJ CHENGAPPA AND SPECTER

release, but the country has not yet got the labelling, infrastructure or mechanism in place. We are in violation of our own rules. It should have been tested to the extent that it could be proven that it is safe. The fact is neither science nor technology operates in a vacuum. Just as science can do a lot good, its application can also do a lot of bad. There is a purity about science, but there may not be purity in the application of science when it turns into technology. It is famously said that the 21st century will be the century of biology. It will be. But after destroying the planet to the extent that we have, we need to make a distinction between science and its application in the form of technology. The difference is human greed. Today, as we see the potential of science and what it has to offer, let us step back with a little modesty. What are the lessons for the application of science? Three words—ethics, regulation and precaution. We must exercise control. The way ahead is progress, but with intelligence, maturity and responsibility; working with the approach that we hold this Earth only in custody for our children.

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## DISCUSSION

**Q. Ms Sahai raised the issue of ethics, regulation and precaution. But Mr Specter you didn't seem to agree with it. Is it really right to give science a free hand without caring too much about?**

**Michael Specter:** If we were to think too much about precaution, regulation or ethics we wouldn't have airplanes, X-rays or vaccines. Precaution is apt, but do keep the greed of a company and the real ability of science to accomplish

things separate.

**Q. Ms Sahai, would Bt brinjal have been more acceptable to you if we did not have a patent issue associated with it? Also, Mr Specter, I am convinced that we will actually have a monster created some day by a genetic mutation. Is there any way of preventing that?**

**Suman Sahai:** Whether Bt brinjal is acceptable to me is immaterial. The question is, has it been

done right? And the answer is "no". The main issue I have is with the technology and very real safety concerns. But that's a matter of science. I think, science and technology must go back to the lab when there are questions. Test it till you are fairly confident that it is safe. If you can't get the wrinkles out, shut the door on it. Also there are socio-political issues involved. There is a very real question of control on seed and the fact that we have a fairly

weak legal framework.

**Specter:** As for preventing genetic mutations, my answer is "no". Any such mutation can destroy the world very easily. But then, come to think of it, we already have enough technology to destroy the world many times over! Safety issues will always exist. But I think we have to draw the line somewhere. Just because someone uses something badly, shouldn't anyone use it ever?